Cover letter

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Editors, npj Systems Biology and Applications

Dear Editors,

Please consider our Article, "Interferon remodeling remains reproducible despite less stable B cell state assignments in systemic lupus erythematosus", for publication in npj Systems Biology and Applications.

Single-cell disease studies often treat annotated states as stable biological units even when cell identity, abundance and transcription differ in reproducibility. We therefore reanalysed public systemic lupus erythematosus datasets using disease-blind B-lineage reconstruction, biological-unit-aware inference, assignment-uncertainty propagation and independent source-label-defined replication.

The central advance is a reproducibility hierarchy rather than a new SLE cell taxonomy. A conventional-B-cell IFN/ISG program reproduced across discovery, internal robustness analyses and an independent GSE135779 cohort despite weak genome-wide concordance. In contrast, end-to-end B_ASC reconstruction failed its prespecified overlap criterion, and corrected source-label-independent external mapping failed the required B_ASC calibration threshold. Regulatory analyses converged on STAT1/STAT2 but remained observational, with broader interferon-response depletion weakening the discovery STAT2 result. The manuscript therefore separates reproducible process-level biology from identity, transfer and mechanistic limits instead of converting those limits into stronger claims.

We believe this evidence hierarchy is relevant to npj Systems Biology and Applications because it addresses a general systems-biology problem in single-cell research: which biological conclusions remain portable when representation, annotation and cohort structure change? The study combines single-cell reconstruction, pseudobulk inference, external replication, calibration analysis and regulatory sensitivity testing in a reproducible framework for systems immunology.

Code, figure source data and the version-specific scientific archive are available at https://github.com/1209433622cz-maker/sle-bcell-remodeling and https://doi.org/10.5281/zenodo.22151739. The work is original and is not under consideration elsewhere. Both authors have approved the scientific content and author contributions. The study received no funding, the authors declare no competing interests, and this secondary analysis used public de-identified data without new participant recruitment or specimen collection. Generative AI assistance is disclosed in the manuscript, with scientific judgement and final decisions remaining author-led.

Thank you for your consideration.

Sincerely,

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